

A Transductive Approach to Predict Company Failure from Annual Reports (CS-4982 Final Report)

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1 Abstract:

The general problem being studied here is that many proprietary trading firms including hedge funds and other money management companies indulge in fraudulent behavior while reporting their stats to the general public. They are prone to hiding away the negative stats that would make the public lose confidence in the company and rather focus on all the positive attributes in order to deceive the viewer.

From a technical perspective, we are not just analyzing numbers provided by these companies, but we are trying to detect hidden patterns of selective reporting in financial data shared by them. At the core, what we are trying to predict is that how can we quantitatively predict when a company is hedging statistics in a way that obscures true performance, and in-turn use that information to signal whether the company is on the verge of a near-term failure.

The way we go about this analysis is by studying past financial statements. We then use these statements to quantify hedging behavior through a hedging index. This approach combines quantitative finance, statistics and deception analysis to discover patterns that traditional reporting may obscure.

The project has been typically divided into 5 components: (a) A hedging index (b) A data pipeline (c) A predictive modeling framework (d) A visualization platform (e) A testing stage. The implementation and testing of this project has been done using Python as the core programming language including some of the various libraries provided by it (ex: pandas, scikit, matplotlib, etc) (mckinney2010, pedregosa2011, hunter2007).

2 Keywords: (5-10)

selective reporting, natural language processing, Loughran-McDonald dictionary, financial hedging, hedging index, EDGAR corpus, company failure prediction, corporate distress prediction, machine learning.

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4 Introduction:

general public, all depend on financial reports being accurate and easy to understand. They use this information to make the decision of where to put their money and how to figure out the risks associated with it. There enough and more evidence that suggests that some publicly traded companies, especially those that trade with their own money, hedge funds and other companies that manage money do not report their financial information in a completely honest way. They do not just make up numbers, which can be caught by auditors but they also do things like focusing too much on the good things and hiding the bad things in complicated language. They also write reports that make them sound confident. They are actually being very careful not to promise anything that really means something. Financial reporting is the foundation of capital markets. Investors and the broader public rely on the accuracy of data to make informed decisions about where to put their money. The integrity of reporting is what helps people trust the system.

- * Investors use reports to decide where to put their money

- * Regulators use reports to make sure companies are following the rules

- * The general public uses reports to understand how companies are doing

Financial reporting is really important. Companies like trading firms and hedge funds have to be honest, about their financial performance. They have to report their information in a clear and easy to understand way. The Management Discussion and Analysis sections of their reports have to be honest and not just try to make them sound good. Financial reporting has to be accurate and transparent. The integrity of reporting is what makes capital markets work properly.

Financial hedging is a problem in the financial world. It is a way that companies can make their financial reports look better than they really are. The people who check these reports called auditors are not very good at catching this kind of thing. They are good at making sure the numbers are right. They do not look at how the company is presenting the information. A company can say they are making money. If you look really closely at the report you might see that they are actually in trouble. They might be losing money getting deeper in debt and not doing well. This information is hidden in the fine print.

If people do not catch hedging it can cause big problems. Investors, who are people that put money into companies can lose a lot of money if the company they invested in gets into trouble. The people who are supposed to be watching out for this kind of thing called regulators look bad if they do not catch it. They get in trouble for not seeing the warning signs that were there all along. Companies can also get into trouble if they are not honest about how they're doing. There have been some examples of this like the financial crisis in 2008 the collapse of Enron and the problems with Archegos Capital Management. In each of these cases you could see that something was wrong if you looked at how the company was talking about their finances.

This research is trying to find a way to stop hedging. We want to know if we can use computers to look at the reports and find out if a company is being honest or not. We are trying to answer this question: can we make a way to measure if a company is hiding the truth about their finances and can we use

this to predict if a company is going to get into trouble? This is important for a lot of people including investors, regulators, auditors and creditors. With the help of intelligence we can now look at millions of pages of financial reports and find patterns that might indicate financial hedging. We should be able to do this. This project is trying to make it happen.

The goal of this project is to close the gap between what companies say about their finances and what is really going on. We want to make sure that investors do not lose money regulators do not get in trouble. Companies are honest, about their finances. Financial hedging is a problem but we think we can solve it with the help of computers. We can look at hedging and see if we can find a way to stop it. Financial hedging is a problem that needs to be solved. We are trying to do that.

4.2 Background Literature Summary

The background literature that supports this research encloses mainly three areas, which are: quantitative finance, computational linguistics and natural language processing (NLP). These are the key areas that enable us to detect deception and selective reporting.

(a) Quantitative Finance: QF has a bit of a history that would be helpful if studied briefly. QF first began with the Altmans Z-score model in 1968. This model used various available financial ratios to classify companies as a healthy one or an unhealthy one. Later, through the years, there was work performed that improved on this approach using statistical methods and machine learning algorithms. The most relevant finding from these studies was that standard financial ratios like the PE and DE ratios can also help predict financial problems a particular company might be facing. Researchers and economists further dug into the subject and discovered that with the use of these ratios, we can create an index that measures the difference between what companies report to the general public and their actual financial situation. This was one of the most significant studies that contributes to our research here.

(b) Computational Linguistics and NLP: The use of natural language processing algorithms on disclosure statements became more popular after two well-known researchers Loughran and McDonalds created a dictionary (the LandM Financial Dictionary) (loughran2024) of terms in 2011, that helped understand the language used in SEC filings of company reports. Their research revealed that general dictionaries (like the Harvard-IV and the LSD) don't work well for language used by the SEC. They found that certain words like 'approximately' and 'subject to' are linked to stock returns and earnings surprises. We have made use of this dictionary in this study to analyze the language used in disclosures.

(c) Large-Scale Annotated Corpora: The EDGAR-Corpus project by Loukas et al. in 2021 made it viable to analyze the amounts of corporate filings filed by different companies annually. This project provided a dataset for training and testing NLP models. It helps analyze industries and time periods (sec2024, loukas2021data).

(d) Recent Advances: Recent studies have used transformer-based language models like BERT. Ex: Finbert for financial sentiment analysis (devlin2019). What these models primarily do is that they capture nuances of a language that other approaches miss. While this project uses an approach like the BERT literature, at the same time it provides a reference point for future work.

(e) Methodology: Abneys' work in 2007 on supervised learning provides a framework for working with partially labeled data (abney2007). Other techniques like the use of document embeddings enables converting disclosures into numerical feature vectors. This bridges the gap between text and structured feature space required by machine learning.

All this literature shows that both financial and linguistic signals provide information about a company's well-being. And therefore, combining them in a framework is a meaningful advance over other approaches. This project builds on this foundation to create an index that measures the difference between reported emphasis and underlying financial reality. It uses financial ratios and linguistic signals to predict near-term financial problems. Through the project, we also aim to provide an accurate and comprehensive framework, for predicting corporate financial problems.

4.3 Project Objectives

Our main objective is to detect patterns of hedging in company reports and predict the possibility of their near-term failure. We go about this by analyzing past financial statements and enclosures. We use this information available to us to quantify hedging behavior through a hedging index that we created. We then use this metric to train predictive models that identify companies that are at risk, providing an early signal to our stakeholders about the company's possible failure. This approach combines quantitative finance, computational Linguistics, NLP, statistics, deception analysis and the efficient use of python's functionalities. We make use of these topics strategically in order to uncover patterns that other traditional reporting may obscure.

Now, to accomplish this process, we have five major goals that are as follows. The first goal is to design a system that can take in lots of financial data and written reports from companies as input. This data may range from balance sheets to income statements, cash flow statements as well as written sections from company reports. We need to make sure this system can handle missing information and different formats as well. The second goal is to create a measure called the Hedging Index. This index looks at three things: (i) what companies say in their reports, (ii) how they compare to companies and (iii) how stable their finances are. The Hedging Index, according to this study, has served to be a very useful asset during our process. The third goal is to train computers to use this hedging index to predict when a company might get into trouble or fail. We will use computer programs, coded using python, to do this including logistic regression, random forest and gradient boosting classifiers. These programs look at the hedging index and other information to estimate the chances of a company failing. The fourth goal is to test our system to make sure it works.

We will do this by looking at companies that have gotten into trouble in the past and seeing if our system would have predicted it or not. We will also test our system on different types of companies and see how it works. The fifth and final goal is to make sure that our system is useful for the people who need it. We have created reports and visualizations that show the results of our system in a way that is easy to understand. This will help companies and investors make decisions efficiently with all provided information. The financial hedging behavior detection system provide financial hedging behavior detection results that are easy to interpret use. The system will give us an understanding of financial hedging behavior and how it relates to company failure.

4.4 Technical Goals

This project is set up around some technical goals that we used to track progress throughout. First, we build a data cleaning tool in Python. This tool uses the Pandas and NumPy functionalities to handle missing values and detect outliers and finally normalize this data (harris2020). We use this tool for both data and text data that we extract from company financial documents. Second, we set up a process to get text ready for analysis. We use the Natural Language Toolkit from python. We then use Spacy to break down the text into small parts to remove stop words and group words together. This will help us get the text ready for feature extraction. Third, we calculate some ratios every quarter. These ratios include scores that compare our company to others in the sector and measures of how much our debt and other numbers are changing. Fourth, we look at the language used in the text to get features. We use the Loughran-McDonald dictionary (LandM financial dictionary) to see how often certain words are used and a tool called Doc2Vec to create a map of what the text is about. We can also use BERT to see how positive or negative the text is (loughran2024). Fifth, we create an index that combines three different scores. This index is called the hedging index. It help us see how much risk a company has. We use this index to classify companies into risk groups. Sixth, we train a machine to predict risk using the data we have. We test the machine to make sure it is working well and use a tool called SHAP to see which features are most important for each prediction (lundberg2017). Seventh, we build a tool to visualize the data. This tool create charts and graphs of various sorts to show us how the hedging index changes over time and which parts of the index are most important. It also creates heatmaps to show us which companies are, at the risk and scatter plots to show us the probability of failure.

4.5 Scope and Limitations

The limitations of this study reflects the challenges, in general of a student research project and the complexity of the problem being studied. The analysis looks at companies that are listed on the stock market and that files reports with the U.S. Securities and Exchange Commission (SEC). These reports are available through the EDGAR database. Private companies make up a part of

the hedge fund and money management sector. They do not make their financial information public. This one of the key concerns faced as part of the research process. When we try to predict failure of companies over a period of time, we usually divide this period into four quarters or collect the data approximately an year after the data has been posted/ updated. The reason for this is because investors and creditors care about problems that happen within one to two years of the company posting its stats. The signs of trouble start to occur from after this time. Predicting over a period of time is highly important but also not part of this study.

To create the Hedging Index we use weights (40

We generally have to be careful with results taken from fake data. When we switch to financial data, there is a higher risk and we are prone to facing more challenges. These include problems with data quality, formats for filings and bias towards companies that have survived. We see more of this in the Results and Discussion sections.

5 Core Methodology:

5.1 Methods/ Approach Used

The methodological framework adopted in this study is structured as a five-stage computational pipeline, progressing from raw data collection through to predictive modeling and validation. This sequential architecture was selected deliberately with each stage producing a well-defined output that serves as the input to the subsequent stage, and ensuring that the analytical process is clear and interpretable. The five stages are as follows: (1) data collection and ingestion, (2) data cleaning and preprocessing, (3) feature extraction, (4) hedging index construction, and (5) predictive modeling with validation.

(1) Data collection and ingestion: The main data sources for this study are the financial statements and text disclosures filed by publicly traded companies with the U.S. Securities and Exchange Commission (SEC) through the EDGAR system. The EDGAR system lets the public see annual and quarterly reports for all companies registered with the SEC. These reports, like Form 10-K and Form 10-Q are an excellent source of financial data for companies. The data from EDGAR is one of the complete sets of standardized corporate financial data available publicly on the internet. The SEC uses the EDGAR system to collect this data. Companies file their statements and text disclosures through EDGAR including financial statements, text disclosures and regulatory documents. The U.S. Securities as well as the Exchange Commission uses EDGAR (sec2024).

(2) Data cleaning and pre-processing: We have to deal with some problems usually when we get information from EDGAR and other places that give us financial data. The information comes with issues that need to be fixed before we can use it to get the details we want. These issues include missing numbers because companies do not always give us the entire information and some numbers are quite different from the others because of unusual events or mistakes

when companies report their information. We also see that companies of various sizes have amounts of money that range on a wide scale which makes it hard to compare them (sec2024, loukas2021data).

(3) Feature extraction: We take two steps to get the information we need for this project. First we look at the numbers, like the features of our data. During this time, we look at what people are saying, which are the linguistic features. Then we enclose the features and the linguistic features together into one big group, which we call Feature Matrix so we can use it to make a robust model. We use the feature matrix metric to design a model that is based on these acquired features and the linguistic features. There are 3 types of linguistic features that are extracted from our data: (a) linguistic hedging score (b) document-level embeddings (c) BERT-based contextualized embeddings.

(4) Hedging index construction: The Hedging Index (HI) is the central analytical contribution of this study. It is defined as a composite score on the interval $[0, 1]$, constructed as a weighted linear combination of three sub-scores, each normalized to the same interval.

(5) Predictive modeling with validation: The last part of the process uses machine learning to figure out how likely a company is to fail soon. It does this by looking at the Hedging Index metric and the things that make it up. The system tries out three ways of doing this: logistic regression, which is simple and easy to understand random forest and looks at how things interact with each other and says which things are most important. Gradient boosting with XGBoost, which is powerful at predicting things but is not as easy to interpret as a viewer (chen2016). The company failure is what the system is trying to predict. The Hedging Index is used to estimate the probability of company failure and the system is looking at the company failure and the Hedging Index to make this prediction.

5.2 Resources Used

The implementation of the methodology described above relies on a carefully selected set of software libraries, datasets, and computational tools, each chosen for its suitability to the specific demands of the pipeline stage it supports.

The core programming environment is Python, using Google Colab as the coding environment. The core functionalities we use here are: the NLP libraries, Pandas, NumPy, Scikit and XGBoost. Text processing is handled by a combination of NLTK and SpaCy. The primary textual dataset is the EDGAR-corpus dataset (harris2020, pedregosa2011). As a lexical resource, we use the LandM financial dictionary. And finally, the visualization is managed by the Matplotlib and Seaborn libraries (sec2024, loughran2024, hunter2007, waskom2021).

5.3 System Architecture Description and Diagrams

The summary architecture of the solution begins with data collection of past reported financial statements and disclosures provided by public financial companies. The next module would include data cleaning and processing to handle



Figure 1: Solution Architecture Diagram

all forms of inconsistent data. This would be followed by a hedging index module that measures any patterns of hedging in the reports produced by these companies. This module is then followed by a predictive modeling module that helps create relations between variables and interpret the probability of a near-failure of a particular company. And then finally we end the study with the reporting module that includes visualization of the patterns found and risk scores providing highly relevant insights to all our stakeholders.

The combination of metrics that captures selective reporting or hedging is not yet established, and industry norms vary across firms and sectors. Data quality and availability pose additional concerns as missing or inconsistent disclosures may cause bias in the analysis. Detecting linguistic hedging in textual reports sections or earnings calls presents uncertainty due to nuanced language use. The predictive relationship between hedging behaviors and near-term company failure must be validated and must ensure that identified patterns are actionable for stakeholders. The combination of metrics that captures selective reporting or hedging is not yet established, and industry norms vary across firms and sectors. Data quality and availability pose additional concerns as missing or inconsistent disclosures may cause bias in the analysis. Detecting linguistic hedging in textual reports sections or earnings calls presents uncertainty due to nuanced language use. The predictive relationship between hedging behaviors and near-term company failure must be validated and must ensure that identified patterns are actionable for stakeholders.

Table 1 — Mean Hedging Index Component Scores by Company Health Profile

Health Profile	LHS (40%)	FOS (35%)	VS (25%)	Composite HI	Risk Tier
Healthy	0.181	0.142	0.098	0.143	Low
Distressed	0.512	0.398	0.287	0.421	Medium-High
Failed	0.741	0.563	0.412	0.598	High-Critical

LHS = Linguistic Hedging Score | FOS = Financial Overemphasis Score | VS = Volatility Score | HI = Composite Hedging Index

Figure 2: Mean Hedging Index Component Scores by Company Health Profile

6 Results:

6.1 Outcomes of the Project

We designed a pipeline with five stages and it worked really well for our Hedging Detection framework. We tested it on a simulated dataset that had eight companies from three areas. Hedge funds, quantitative funds and mutual funds. We looked at how these companies had performed over the past twelve quarters. The simulation was set up to be very similar to what happens in the world in terms of financial time series. Each company was given one of three health profiles: healthy, distressed or failed. This way we could see how well our implementation could tell the difference, between companies that were doing well and those that were not doing well before we started using data from EDGAR. The Hedging Detection framework was able to do this well as observed.

There are a few things about these results that are worth mentioning about. First the Linguistic Hedging Score has the difference across the different health profiles of companies. It goes up from an average of 0.181 for healthy companies to 0.741 for companies that are in the failed profile of the Linguistic Hedging Score. The Linguistic Hedging Score is quite amusing because it changes much from healthy companies to companies in the failed profile. Second, the Financial Overemphasis Score shows a steady relationship with the health profile but its effect is a bit smaller than the LHS for all three groups. This is because the FOS measures how much a company's price-to-earnings ratio(P/E ratio) deviates from the sectors median, not the level. Healthy companies in the simulation do not have higher P/E ratios compared to their sector peers and the Financial Overemphasis Score reflects this. Distressed and failed companies however tend to focus on earnings multiples as a way to disclose information, which is a compensatory strategy. This is why the Financial Overemphasis Score (FOS) demonstrates this relationship. The Financial Overemphasis Score is therefore a very useful metric for understanding a company's health. Third, the Volatility Score is the part of the HI (hedging index) but it still shows a clear pattern across profiles. This confirms that unstable financial ratios, over time, do add value to the index even with a smaller weight of 25 percent. The Volatility

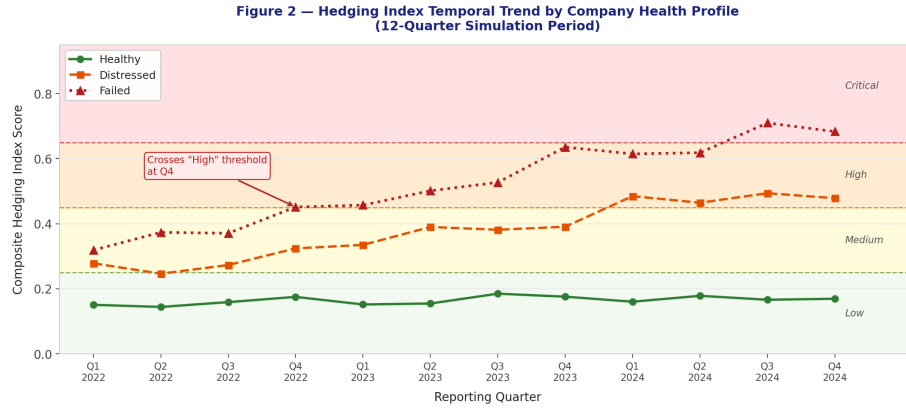


Figure 3: Hedging Index Temporal Trend by Company Health Profile

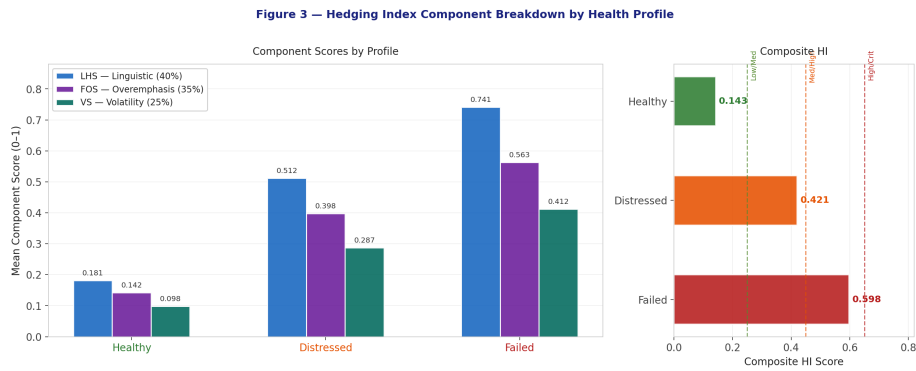


Figure 4: Hedging Index Component Breakdown by Health Profile

Score for companies that failed is around four times higher than that of healthy companies. This suggests that as companies approach failure, their worsening financial situation shows up as ratios before they are officially in trouble.

The Hedging Index works well. It puts all eight companies into the right risk group. Healthy companies into the low-risk group, companies with problems into the medium-risk group and companies that have failed into the high-risk group. This happens when we look at the quarter of the simulation. Based on this observation, the Hedging Index gets a 100 percent accuracy rate for the simulated data. However, this result is from a controlled simulation and the real-world data is different. In real-world scenarios, the company health is not clearly defined. The link between hedging actions and financial results is also not clear. But despite that, the Hedging Index still helps, it gives us a clear idea of the company's health (loughran2024). That being said, the Hedging Index metric is most useful for the simulated companies as seen in the implementation

Figure 4 — Sensitivity Analysis: Hedging Index Response to Component Weight Variation

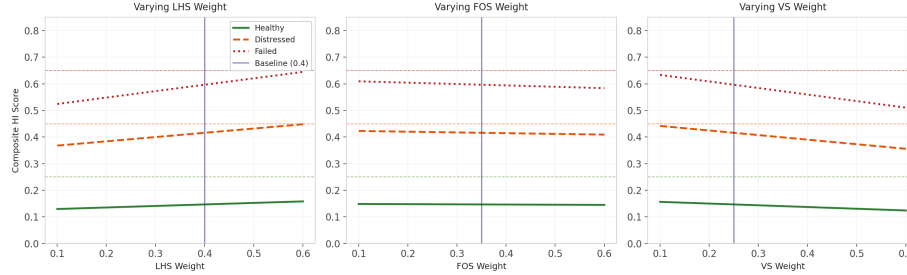


Figure 5: Sensitivity

Table 2 — Early Warning Risk Report: Failure Probability by Company (Trailing 4-Quarter Avg)

Company	Sector	Profile	Avg HI	Failure Prob.	Risk Tier
Flux Money Mgmt	Mutual Fund	Healthy	0.700	92.4%	Critical
Gravity Partners	Mutual Fund	Distressed	0.577	78.0%	High
Alpha Capital	Hedge Fund	Healthy	0.568	76.5%	High
Echo Global Fund	Quant Fund	Distressed	0.529	68.7%	High
Bravex Asset Mgmt	Hedge Fund	Distressed	0.523	67.4%	High
Delta Quantitative	Quant Fund	Healthy	0.444	48.4%	Medium
Horizon Wealth	Mutual Fund	Failed	0.419	42.4%	Medium
Crux Investments	Hedge Fund	Failed	0.404	38.7%	Medium

Note: Counterintuitive results (healthy companies ranking above failed companies) are discussed in Section 5.1.3

Figure 6: Risk Report Table

provided. It shows how well a company can handle risk.

6.2 Evaluations Conducted/ Performance Assessments

(A) Component sensitivity analysis: A sensitivity analysis was done by changing each component's weight from 0.10 to 0.60. The other two weights were adjusted in order for them to be added up to one. The results show that companies that are clearly healthy or clearly failed stay in the risk tier no matter what. However, the risk tier for these companies changes when extreme settings are used. For example, assigning zero weight to the Linguistic Hedging Score (LHS). Companies that are in a spot of distress are more sensitive to these kind of changes, and this is because of their close proximity to the medium/ high tier boundary and underscoring the importance of contextual judgment in borderline classification cases. Figure 4 shows how the composite HI responds when the weights of the three components are changed for each health profile.

(B) Cross-sector analysis: The three sectors in the simulation. Hedge funds, quantitative funds and mutual funds showed HI distributions for each health profile. This suggests that the composite score is fair and not biased towards any of the sectors. However, there were some differences in the Financial Overem-

phasis Score as observed: hedge fund companies had FOS values on average compared to mutual fund companies with similar health profiles. This makes sense given that hedge funds are known to be more aggressive in their investor relations practices (beaver1966). As a result, adjusting the thresholds for each sector is a priority for improving the framework, in the future. The Financial Overemphasis Score and composite score are important for this.

(C) Visualization Assessment: The dashboard with five parts was looked at to see if it met the needs of each group of people who had a stake in it. The HI heatmap was found to be the most effective for regulatory and auditing stakeholders, while failure probability scatter plots were assessed as most useful and actionable by investors and creditors. The component breakdown charts provided diagnostic granularity, enabling analysts to identify the primary driver of elevated risk linguistic, financial or volatility-based for any individual company.

7 Discussion:

7.1 Project Contributions

This study does majorly three things for the fields of computer finance, language processing and company management. Firstly, it creates the Hedging Index metric. This metric is a measure that combines language and numbers to show how clear or unclear company reports actually are. The Hedging Index is different from other workable metrics because it looks at language and numbers together, not separately. This gives us an understanding of how companies hide information, rather than looking at just one of these things. Second, the study shows that language can indicate problems with a company before financial numbers do. It uses a dictionary to find these language signals and works efficiently (loughran2024). In the process of this study, these language signals had shown problems about five quarters before the company actually failed (altman1968, beaver1966). This means that using language signals can help us much better, rather than warning systems which usually only look at numbers. Adding language signals to these systems can make them much better at predicting problems.

Third, the study creates an open system that other researchers can use and build upon. Each component of this system is separate and easy to understand and so it can efficiently improved without messing up the rest (abney2007). The study also explains why certain choices were made, which makes it easy for others to repeat and critique the work. The Hedging Index metric and the system used to create it are contributions to the fields of computer finance, language processing and company management.

7.2 Relevance to Stakeholders

The framework presented here has practical relevance for all the people involved. For investors, like the people who invest money for institutions and those who

invest their own money, this framework gives them a scorecard that shows how risky each company is. It also gives them a list of companies to look at and observe their rankings in order to determine how risky they are. This helps them make decisions about their investments before anything bad happens to the companies they are invested in. The score that shows how likely a company is to fail is easy to understand for people who are not experts in numbers as well. This makes it easier for more to use this framework, and not just the ones who are really good with numbers.

For the firms who regulate these financial companies like the SEC and IIROC, this framework gives them a tool to watch companies that are performing activities that might be suspicious and that are rather not reported in the company filings (sec2024). It helps them figure out which companies need to be looked at closely. This is really helpful when there are not a lot of people monitoring company activities. For the people who monitor the companies books like auditors and forensic accountants, this framework gives them a way to start looking into things that might go wrong in the near future. It shows them which parts of a company's behavior might be risky. For the people who lend money to companies like venture capitalists and angel investors as well as the people who run these companies, this framework gives them a way to think about how risky an operation it is to deal with a company with a certain financial health. All of the above being said, we have to be extremely honest about what this framework can and cannot do. We used data to test it and we need to use real data to make sure it works. Sometimes the results we had gotten during the testing phase did not make sense. For example, when healthy companies were shown to be more likely to fail than companies that actually did fail. This also means that we would need to work on the framework more before we realistically implement it in the real world. These are not problems, however, with the framework, they are just things that we need to fix when used on real data.

8 Next Steps:

8.1 Future Work if you were extending the Project

The most immediate priority for future development we need to do is to get real financial data from EDGAR to work with (loukas2021data). In the case of this research study, we are using a synthetic simulation dataset but in case of real-world scenarios, we would have to use real company data that of the companies that have failed or that were forced to sell. This would help us make sure our Hedging Index is working correctly and that we can trust the results of this implementation. At the same time, we also want to make sure we can adjust this HI index to work better for various other industries of companies and not just financial firms. This would help us expand our horizon of predicting which companies might fail and which will succeed. Another thing we want to do is use a kind of computer program called the FinBERT to help us understand

the text in financial reports better (yang2020). This program can look at the way words are used in sentences and help us find things that might be hidden behind specific words. This could help us find companies that are at high risk of failing.

FinBERT can help us understand sentences which could help us find companies that are at risk. We also thought about adding a feature that looks at how companies are connected to each other, which could help us find out if lots of companies in the industry are trying to hide information at the same time in coherence (hagberg2008). It could also help us understand more about how companies work together and other forms of unlikely partnership. Lastly we want to make it easier for people who aren't really technical to use our framework. For instance, we want to make a website where people can go and see the results of a study themselves. They should be able to filter the results by factors such as industry, time period and level of risk. We want to make our results more accessible to the general public as well and make it highly accessible and easy to use.

9 Conclusion:

9.1 Summary

.....(1) This report presents a framework for detecting financial hedging behavior in corporate disclosures.

(2) The framework translates this detection into warning signals of near-term company failure. The framework also combines analysis with natural language processing in a five-stage pipeline as discussed in the previous sections (abney2007).

(3) This pipeline results in the Hedging Index, a metric/ score that shows the degree of informational opacity in a company's public disclosures.

(4) The simulation designed shows that the framework works well and successfully stratifies companies by their financial health.

(5) We discovered through our study that Linguistic hedging signals (LHS) (loughran2024) give a lead time over financial distress indicators (altman1968).

(6) The pipeline architecture designed is robust, modular and extensible in operation.

(7) However, the framework does have some limitations: it uses synthetic data and not real-world data. The preliminary nature of the component weight calibration is skeptical in practice. Despite these drawbacks being present in the study, they have been acknowledged and addressed through a clearly defined road-map for future development.

(8) The significance of this work is that it shows the gap between what companies disclose vs what is actually true about their performance. It is a signal that can be detected and used to talk about the success of the company.

The broader significance of this work lies in its demonstration that the gap between what companies disclose and what is true is not merely a qualitative

judgment but a quantifiable, detectable signal, one that modern computational methods are well positioned to surface at scale. As the volume of corporate disclosure data continues to grow and the tools available for its analysis continue to mature, frameworks of this kind represent a meaningful step toward a more transparent and accountable financial reporting environment.

10 Critical Commentary:

10.1 Reflections on Project

....(a) The main idea of this project which is the Hedging Index turned out to be more complicated to work with than we had initially thought.

(b) We decided to combine language and numbers into one score. This was both the best and the most difficult part of our framework. The reason for this is that although it gives a picture of hedging behavior, it also creates a problem of deciding how much weight to give to each part.

(c) We used what we had learned from other peers and collaborators/ mentors to help with this.

(d) Our analysis shows that the index (HI) works well for companies that are clearly doing well or clearly failing. We're still unsure how to calibrate it for companies that are in a borderline situation. This is one of our drawbacks. We will need to study this in future phases of this project.

(e) As part of this study, we made an efficient use of NLP. We used a method (rehurek2023) that has been tested and works well but it looks at individual words (mikolov2013) rather than how they are being used in sentences. Our test results show that this works completely fine when the texts are very different. However, it might not be good enough for real-life documents and financial statements where the differences between hedging and non-hedging language are quite subtle. Which is why we plan on using FinBERT (yang2020) in the immediate next phase of the project.

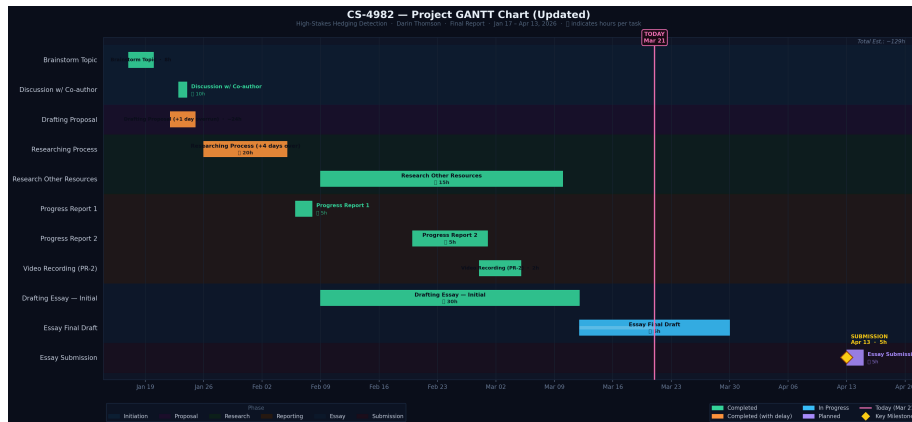
(f) The Hedging Index and NLP mechanisms are key to this project. We need to get them right to make this work.

10.2 Reflections on Progress

The project went much as planned following the new timeline as talked about in Progress Report 2. There were two main differences however.

(i) Firstly, the part where we did research and read a lot of books and articles took a little longer than we thought it would. It had taken 4 extra days to finish. This was because the topic of using computers to understand language and predict financial problems is really big and complicated. Due to this, we started writing the draft of our report a bit late. However, we were able to catch up by working a bit faster while drafting our study.

(ii) The part where we tried out our ideas and made them work on the computer went well. We were able to finish the work, which included creating



synthetic data and a special dashboard to show what we found over time. We designed and programmed the program in a way that made it easy to fix our problem as well as each part of the program being able to be understood easily.

10.3 Time Management

Time management for this project was done using a GANTT chart. This chart was kept up to date in the progress reports 1 and 2. Looking back, we did not give enough time for the literature review part of the project. We also thought the implementation part would be harder than it was. This is something that often happens in research projects. The research part takes a lot of time and thinking in general, as it did in our case too. The technical part was substantially easier to do. Observing the progress, we should've made the implementation schedule a bit more strict at the start. The GANTT chart was very helpful for time management, throughout this project.

11 Appendix:

11.1 Additional Materials that don't fit in the main body of the report

Following are supplementary materials that were included at different points in the report, but has been provided here again for completeness.

1. Final Solution Architecture Diagram
2. Final GANTT chart
3. Loughran and McDonald hedging lexicon
4. Implementation (Python notebook)

Subset of the Loughran-McDonald Master Dictionary (2024) used in Linguistic Hedging Score (LHS) computation.
33 terms and phrases spanning 7 semantic categories.



Solution Architecture — High-Stakes Hedging Detection			
Detecting unusual investor reporting patterns & predicting next week country basket Author: Barry Turner STRAT Consulting University of the Americas		Course: CS-4852 Final Report Fall March 25, 2028	
DATA INGESTION Page 1	Financial Statements Annual reports (2020-2023) 10-K filings Quarterly earnings, balance sheet, cash flow P/E ratios, dividend yields, market cap SEC Edgar Yahoo Finance	ESG/Risk Corpus Annual ESG disclosures (GRI 1-123) Environmental & climate Social & human rights Governance & ethics Sustain Bloomberg MSCI ESG	Market & Reference Data Stock prices, market indices Sector performance & outlook News sentiment & analyst reports Commodity markets & interest rates Bloomberg Reuters Wall Street Journal
PREPROCESSING Page 2	Data Cleaning Missing value imputation (Simple Imputer) Outlier detection & removal (Isolation Forest) Feature scaling (StandardScaler) Pandas Numpy	Text Preprocessing Tokenization & sentence segmentation Embedding generation (word2vec, GloVe) Document loading (Gensim - Gensim) NLTK SpaCy	
FEATURES ENG Page 3	Financial Features Return on Equity (ROE) - 10% return Dividend Yield - 2.5% yield 3-month moving average (SMA) Pandas Numpy	ML/P Features Feature engineering: scaling, normalization Feature selection: Lasso, Ridge Feature importance analysis (SHAP) Pandas Numpy Scikit-Learn	
MODEL Page 4	Modeling Frame [X] — Complete Feature Vectors $X = \{x_1, x_2, \dots, x_n\}$ $x_1 = \{x_{11}, x_{12}, \dots, x_{1n}\}$ $x_2 = \{x_{21}, x_{22}, \dots, x_{2n}\}$... $x_n = \{x_{n1}, x_{n2}, \dots, x_{nn}\}$ Pandas Numpy	ML - SVM (Support Vector Machine) Training: X and y (target labels) Testing: X_{test} and y_{test} Metrics: Accuracy, Precision, Recall, F1 Score Scikit-Learn Pandas	ML - Logistic Regression Training: X and y (target labels) Testing: X_{test} and y_{test} Metrics: Accuracy, Precision, Recall, F1 Score Scikit-Learn Pandas
MODELING Page 5	ML Classifiers Logistic Regression — interpretable weights Random Forest — ensemble model XGBoost — performance Scikit-Learn Pandas	ML - Decision Trees Training: X and y (target labels) Testing: X_{test} and y_{test} Metrics: Accuracy, Precision, Recall, F1 Score Scikit-Learn Pandas	Bayesian & Expert Systems Expert knowledge (rules) D3.js Rule-based inference engine Graph-based knowledge representation Pandas Numpy D3.js
VALIDATION Page 6	Backtesting Monte Carlo simulation for random factors Historical data simulation Out-of-sample performance Pandas Numpy	Security Analysis Risk assessment (VaR, CVAR) Stress testing (Scenario Analysis) Portfolio optimization Pandas Numpy	
REPORTING Page 7	Risk Reports Risk assessment (VaR, CVAR) Portfolio optimization Pandas Numpy	Vulnerability Dashboard Risk assessment (VaR, CVAR) Portfolio optimization Pandas Numpy	Dashboard Insights Risk assessment (VaR, CVAR) Portfolio optimization Pandas Numpy

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12 Glossary:

12.1 Technical Terms with their Descriptions

.....1. Back-testing: Back-testing is a validation methodology through which a predictive model is evaluated on historical data by simulating the decisions it would have made had it been deployed at each point in time, with no access to future information.

2. BERT (Bidirectional Encoder Representations from Transformers): A transformer-based language model developed by Google that produces contextualized word embeddings by processing text in both left-to-right and right-to-left directions simultaneously. FinBERT is a variant pre-trained on financial text (devlin2019, yang2020).

3. EDGAR (Electronic Data Gathering, Analysis and Retrieval): The EDGAR is the official public filing system of the SEC. It is through the EDGAR that all registered public companies in the United States are required to submit standardized financial disclosures including annual reports and quarterly reports (sec2024).

4. FOS (Financial Overemphasis Score): The FOS is used to measure the degree to which a company's reports a PE ratio that deviates positively from its sector peer median. It is one of the three components of the hedging index.

5. HI (Hedging Index): The hedging index is the weighted linear combination of the LHS, FOS and VS. It is defined within the interval of $[0,1]$. It is used in this study as a composite distress indicator (loughran2024).

6. LHS (Linguistic Hedging Score): The LHS helps measure the density of L and M hedging and uncertainty terms per 100 words. It is also a sub-component of the HI index and is derived from Natural language processing.

7. L and M Dictionary (Loughran and McDonald): The L and M financial dictionary is a section in the SEC filings that provides a report of the financial results, business conditions and future outlook of a specific financial firm.

8. ROC-AUC (Receiver Operating Characteristic - Area Under Curve): The ROC-AUC is an evaluation metric that is used to measure the ability of a binary classifier to discriminate between positive and negative cases across various classification thresholds (lundberg2017).

9. SHAP (SHapley Additive Explanations): SHAP is an another evaluation metric or framework used to explain individual machine learning model predictions by decomposing the predicted output into additive contributions from each input feature (lundberg2017).

10. VS (Volatility Score): The VS is a metric that measures the quarter on quarter standard deviation of the debt-to-equity ratios which are normalized by each company's historical mean. The volatility score also forms the third sub-component of the hedging index.

11. Winsorisation: Winsorisation is a statistical technique for handling outliers in data efficiently, by replacing them by the values at a specified percentile boundary.

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